

Shyam Kumar

Patna, Bihar

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GitHub

Education

Birla Institute of Technology, Mesra

Bachelor of Technology (Electrical and Electronics Engineering) CGPA - 8.42

October 2023 – Present

Ranchi, Jharkhand

Vidya Niketan High School

12th – 80.8%

May 2020 – 2022

Patna, Bihar

Skills

Technical: Pandas, Python, Embedded C.

Core: Power Electronics, Electrical Machines, Control Theory, Power system.

Tools: MATLAB, Simulink, NI Multisim, Power BI.

Interpersonal: Problem Solving, Analytical skills, Leadership, Event Management, Teamwork, Effective Communication.

Projects

AI-Based Battery Parameter Estimation | ANN, Simulink, Arduino, MKR Memory Shield

- Developed a **real-time AI-based battery health monitoring system** using sensor data acquired from embedded hardware.
- Designed and trained an **ANN model** to estimate intrinsic battery parameters including **internal resistance, capacitance, hysteresis, and diffusion**.
- Validated and optimized the estimation framework using **Simulink**, followed by hardware-level testing on **Arduino** for real-world performance.

Industrial Safety and Surveillance System | Python, OpenCV, YOLOv9, PySide6

- Developed an **industrial safety and surveillance system** for real-time monitoring of personnel and restricted areas using CCTV feeds.
- Integrated **YOLOv9-based object detection** to identify persons, safety equipment, and hazardous objects, enabling rule-based safety checks.
- Built a **Desktop-Monitoring Dashboard** using Pyside6 to display live video streams and generate safety alerts.

Stocks Comparison & Prediction Model | Power BI, DAX, Excel, Pandas

- Developed a **Power BI stock analytics and forecasting dashboard** to compare performance across five major companies.
- Implemented **SMA crossover trends, volume-return correlations, and built-in forecasting** to analyze and predict closing prices with confidence intervals.
- Designed an **interactive dashboard** with buy/sell markers, YTD return tracking, and interactive visuals — enhancing investment insight and comparison accuracy by 45%.

Achievements

- 1st Rank** in Summer Project Program among **50+ students** conducted by EESoc Student Branch-BIT Mesra.
- 1st Rank** in Project of Battery Management System among **25 teams** organized by EEE BIT Mesra.
- Finalist** in **Code for Bharat** Hackathon among 1000+ teams conducted by Tech Masters India.
- Runner-up** in the Bot Soccer event organized by BIT Mesra.
- Organized **Hack-a-BIT** for students of BIT Mesra.

Positions of Responsibility

Placement Coordinator-BIT Mesra

April 2025 – Present

- Successfully contributed to **placement drives** for 10+ companies, resulting in improved overall placement statistics.
- Organized multiple **career guidance** and **mentorship sessions**, enhancing students' career awareness and focus.

Tech Head-BIT Mesra

March 2024 – Present

- Organized various **software and electronics domain** events, enhancing students' technical and practical skills.
- Coordinated with **faculty and industry mentors** to plan and execute impactful technical initiatives.